Final Project Report

ENGR3540 Introduction to Feedback Control

Project Title: Implementation of a Closed-Loop Web Tension Regulation System

Name of Students: David A. Williams

Date of Submission: 05-02-2026

Name of Instructor: Dr. Lei Miao

Department of Engineering Technology

Middle Tennessee State University

Project Topic and Description

The objective of this project is to design and implement an industrial based Long-Loop tension control system. The system manages the slack (arch depth) of tape between two independent motor-driven spools: a Master (Winder) and a Slave (Unwinder). Were the Winder and Unwinder are to demonstrate two independent processes, were the slack between the two needs to maintain a certain “buffer” between the two.

In industrial automation, material tension must remain constant to prevent stretching or breaking. This system utilizes a high-speed Laser Distance Sensor to measure the depth of the slack loop as the Process Variable (PV). A Proportional-Integral-Derivative (PID) controller then adjusts the Slave motor speed to maintain a target depth (Setpoint), effectively rejecting disturbances caused by Master motor velocity changes.

Design

1. Hardware Architecture

- Controller: Mitsubishi FX5U-32MT/DSS. Selected for its built-in Analog-to-Digital converters and High-Speed Transistor outputs, which are required for Pulse Train Output to control stepper motors.
- Actuators: Two STP-MTRD-17038 Integrated Stepper Motors. Controlled via the PLSY instruction (Pulse/Direction).
- Sensor: Wenglor OPT2011 Laser Distance sensor. Configured for 0-10V DC analog output to represent a physical depth range of 250mm to 600mm.

2. Signal Conditioning

- Per the techniques learned in Lab 7, manual scaling was implemented in the PLC to ensure PID registers use physical units, mm, rather than raw digital counts, 0-4000.

  Scaling Formula: D502 := ((Analog_Value / 4000.0) * 350.0) + 250.0

3. Mathematical System Model

The system was modeled in the Laplace domain as a second-order system with an integrator.

- Plant Model Gp(s): The slack loop acts as a physical integrator (1/s). To achieve high accuracy, Actuator Lag (tau = 0.15s) was added to account for motor inertia.
- Transfer Function: Gp(s) = 0.075 / [s(0.15s + 1)]
- System Type: This is a Type 1 System. According to Lecture 7, a Type 1 system provides zero steady-state error (ess=0) for step inputs, ensuring the arm always returns to the exact setpoint.

4. Controller Implementation

The control logic utilizes a Parallel PID algorithm.

- Forward Loop: Slave_Speed = PID_Output.
- Advanced Logic: A "Rewind/Reverse" bit was implemented that simultaneously flips the physical motor direction and swaps the PID roles (Master <-> Follower), ensuring the negative feedback loop remains stable in both winding directions.

Tuning and Testing

1. Stability Analysis:

Using MATLAB, a high-resolution Root Locus was generated. The analysis showed a stability "circle" near the origin.

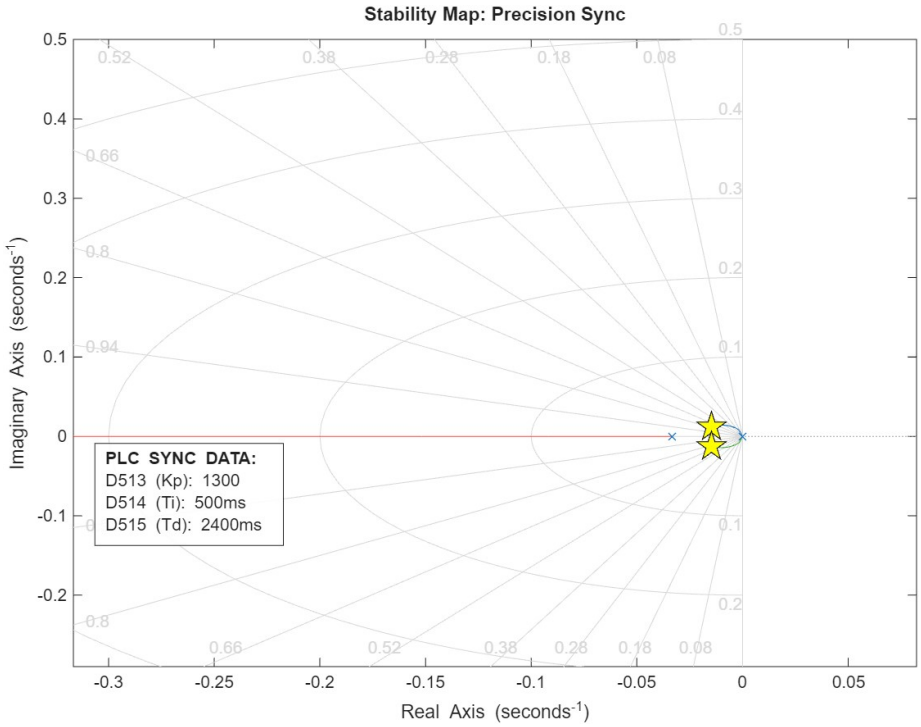


Figure 1: Root Locus Stability Map. The gold stars indicate the closed-loop pole locations for the final tuned parameters (Kp=13, Ti=500ms, Td=2400ms). The placement in the Left Half Plane (LHP) confirms the system is asymptotically stable.

- Pole Placement: The Gold Star on the Root Locus plot confirms that the closed-loop poles are located deep in the LHP.
- Damping: The tuning resulted in a high damping ratio (zeta > 0.8), preventing underdamped oscillations ("hunting") that could damage the web material.

2. PLC Unit Synchronization:

A critical calibration step involved matching the FX5U's specific time units. Synchronizing these units between the PLC and MATLAB allowed for a "Digital Twin" simulation that accurately predicted physical behavior.

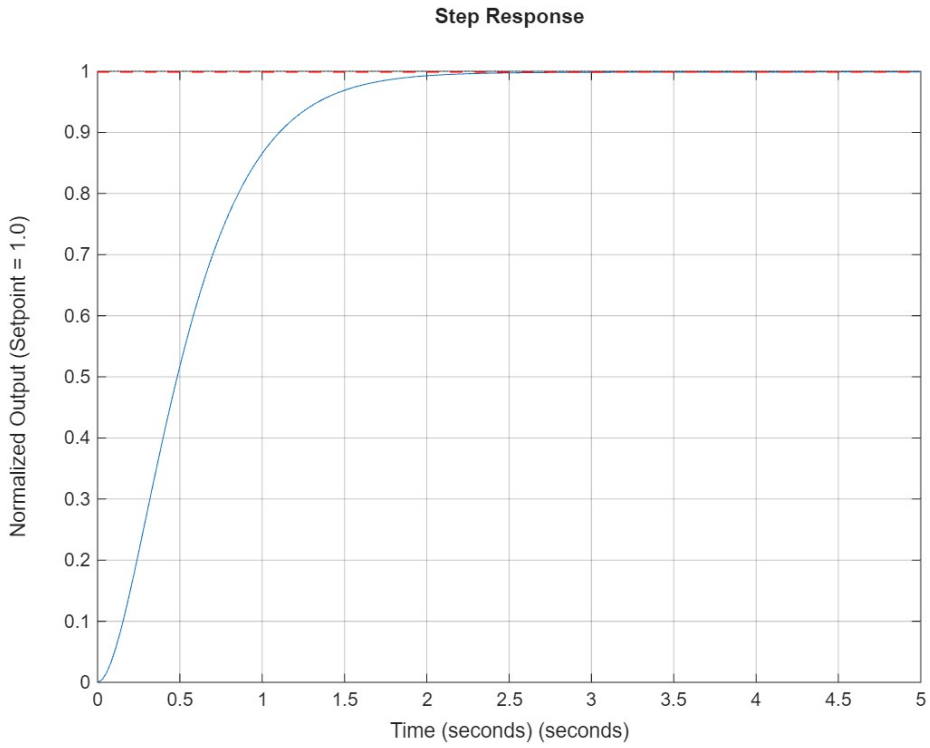
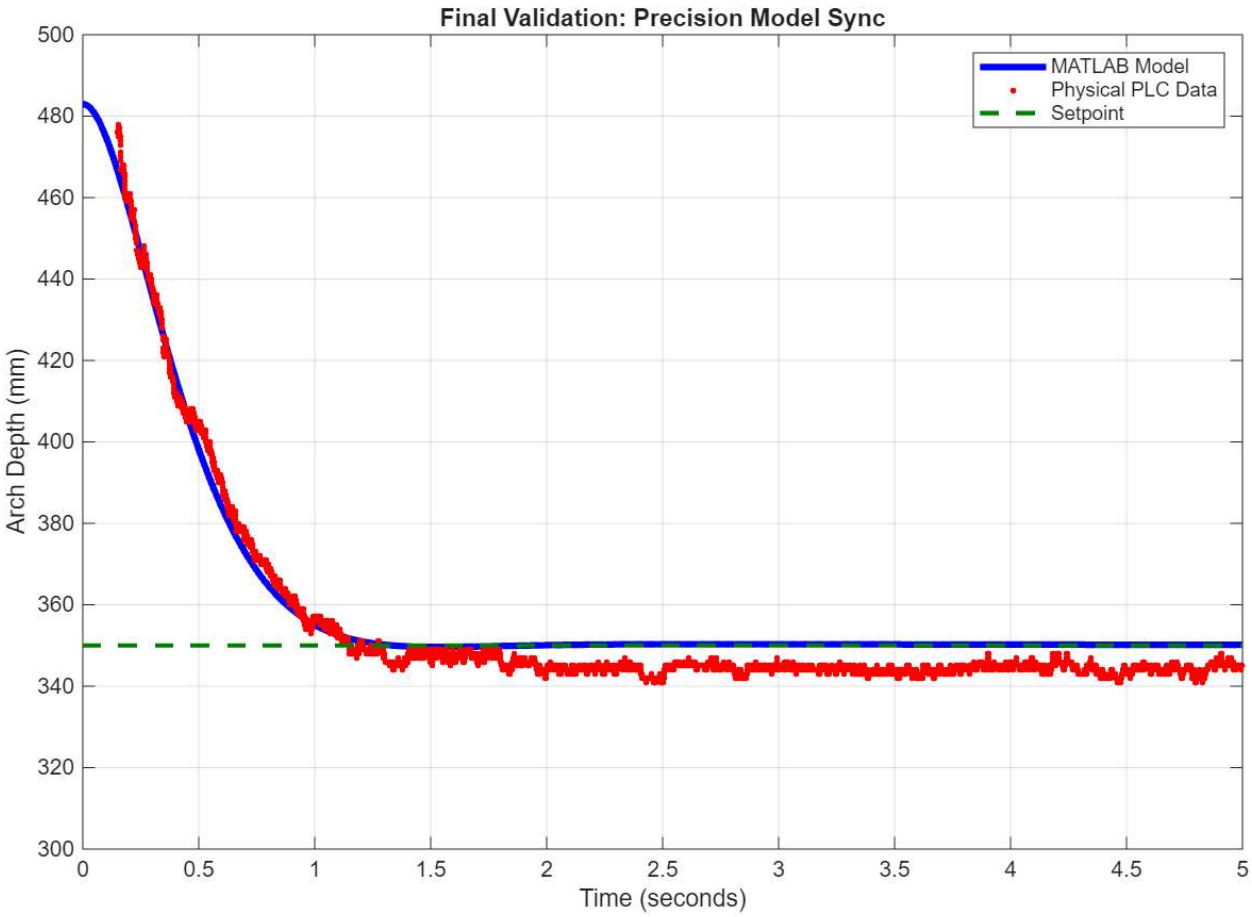


Figure 2: Theoretical Unit Step Response. The blue curve shows the system's transition to the normalized setpoint (Red Dashed Line). The smooth "S-curve" reflects the inclusion of motor lag and derivative damping, providing a 1.5-second rise time with zero overshoot.

3. Model Validation:

The final validation involved exporting physical sensor data from GX LogViewer and overlaying it on the MATLAB simulation.



Conclusion

In this project, the principles of feedback control, Stability (Ch 6), Steady-State Error (Ch 7), Root Locus (Ch 8), and Design via Root Locus (Ch 9), were successfully applied to a physical mechatronics system.

The transition from Version 1.0 (Simple Follower) to Version 2.0 (Pure Regulator) demonstrated the power of the Integral term in "discovering" unknown velocities. The successful validation against physical data proves that mathematical modeling, when properly calibrated for hardware lag and non-standard units, provides a reliable roadmap for industrial system tuning. The final system is stable, accurate, and robust enough to handle variable line speeds and bidirectional operation.